V3 empirical study · Instruction

Six models · three underlyings · four volatility regimes · 1.5-year monthly calibration

What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of six models tracks realized stock price S_t most closely, and does the ranking change across companies and the four volatility regimes?
- This report does not price options. Option RMSE lives in the companion decision PDF.
- In every table the BEST row is the lowest percentage RMSE of the p50 path versus realized S_t. That row is bold and shaded green; Mark ranks 1–6.

Experimental design (held fixed for every model)

- Models: GBM, GARCH(1,1), Heston (no jumps), Merton jump-diffusion, GARCH–Merton, Heston–Merton (Bates).
- Companies: SPY, AAPL, MSFT. Each name is calibrated and simulated separately.
- Regimes: Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- Calibration window: 1.5-year ending at each monthly update. Rolling: monthly. No look-ahead.
- P-measure: keep the lookback drift μ. Do not replace μ → r (that replacement is only for LSM option pricing).

V3 empirical study · Instruction (continued)

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Path construction

- One rolling path cloud per ticker × regime × model, n_paths = 10000, seed 42.
- Clock: daily trading days from the first bar of the regime through the regime end. $\Delta t = 1/252$.
- Start at the observed open S_0. Refresh parameters at month-ends with data ≤ that date.
- At each day, summarize the cloud by p10, p25, p50, p75, p90. The reported path is p50, not the Monte Carlo mean.

Metrics

- Primary: RMSE% = $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. Ranking uses this only.
- MAE = mean $|p50 - S_t|$. Bias \$ = mean$(p50 - S_t)$. Bias% = $100 \times \text{mean}((p50 - S_t)/S_t)$.
- ICP 25–75 = share of S_t inside [p25, p75]. ABW = mean(p75 − p25). ICP 10–90 is recorded but not a table column.
- ICP is higher-better and is not used for the green row.

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t. 1.5-year lookback, monthly rolling, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
Modified GBM	**28.81**	**20.4361**	**+19.9382**	**+23.42**	**34.9%**	**24.7230**	**1**

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
Modified GBM	**3.93**	**6.2957**	**+6.2878**	**+3.25**	**87.4%**	**20.8232**	**1**

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
Modified GBM	**6.73**	**13.2464**	**+10.3903**	**+3.96**	**63.7%**	**35.5580**	**1**

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
Modified GBM	**8.90**	**22.2581**	**-13.2944**	**-3.73**	**54.0%**	**46.8858**	**1**

RMSE% is $100\times\sqrt{\text{mean}(((p50-S_t)/S_t)^2)}$. MAE and Bias $ are in stock-price dollars. ICP 25–75 = share of S_t inside [p25, p75]. ABW = mean(p75−p25). Ranking uses RMSE% only.

Page 4 · Table block · AAPL · six models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t. 1.5-year lookback, monthly rolling, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
Modified GBM	**48.93**	**1.1596**	**+0.9616**	**+34.47**	**38.1%**	**1.8180**	**1**

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
Modified GBM	**23.70**	**4.4831**	**-4.3687**	**-19.47**	**26.9%**	**3.9409**	**1**

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
Modified GBM	**30.74**	**11.6527**	**+11.6416**	**+26.80**	**16.7%**	**13.3953**	**1**

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
Modified GBM	**29.10**	**21.5615**	**-21.5615**	**-26.30**	**5.2%**	**14.7813**	**1**

RMSE% is 100×√mean(((p50−S_t)/S_t)²). MAE and Bias $ are in stock-price dollars. ICP 25–75 = share of S_t inside [p25, p75]. ABW = mean(p75−p25). Ranking uses RMSE% only.

Page 5 · Table block · MSFT · six models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t. 1.5-year lookback, monthly rolling, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
Modified GBM	**20.88**	**2.5199**	**+1.4998**	**+11.74**	**47.2%**	**5.5069**	**1**

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
Modified GBM	**8.46**	**2.8349**	**-2.6665**	**-6.91**	**83.8%**	**7.2140**	**1**

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
Modified GBM	**15.52**	**16.3571**	**+16.3571**	**+14.59**	**36.7%**	**26.9838**	**1**

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
Modified GBM	**8.56**	**12.4401**	**-10.3224**	**-5.46**	**90.1%**	**36.6256**	**1**

RMSE% is 100×√mean(((p50−S_t)/S_t)²). MAE and Bias $ are in stock-price dollars. ICP 25–75 = share of S_t inside [p25, p75]. ABW = mean(p75−p25). Ranking uses RMSE% only.

Page 6 · Table block · AMZN · six models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t. 1.5-year lookback, monthly rolling, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
Modified GBM	**60.91**	**1.2595**	**+1.2311**	**+44.54**	**46.4%**	**2.1719**	**1**

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
Modified GBM	**26.98**	**4.0209**	**+4.0199**	**+25.00**	**8.3%**	**5.2912**	**1**

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
Modified GBM	**45.06**	**37.8664**	**+37.8664**	**+43.08**	**0.4%**	**37.9814**	**1**

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
Modified GBM	**18.37**	**17.9199**	**-15.7442**	**-10.92**	**60.3%**	**27.9372**	**1**

RMSE% is 100×√mean(((p50−S_t)/S_t)²). MAE and Bias $ are in stock-price dollars. ICP 25–75 = share of S_t inside [p25, p75]. ABW = mean(p75−p25). Ranking uses RMSE% only.

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells bes
1	Modified GBM	24.10	22.29	16

Table S2 · Wins by company (lowest RMSE% in that company×regi

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
Modified GBM	4	4	4	4	16

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
Modified GBM	39.88	15.77	24.51	16.23	24.10

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Page 8 · Summary by company · mean RMSE% across four regimes

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
Modified GBM	**12.09**	**7.82**	**4**

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
Modified GBM	**33.12**	**29.92**	**4**

Table S4.3 · MSFT

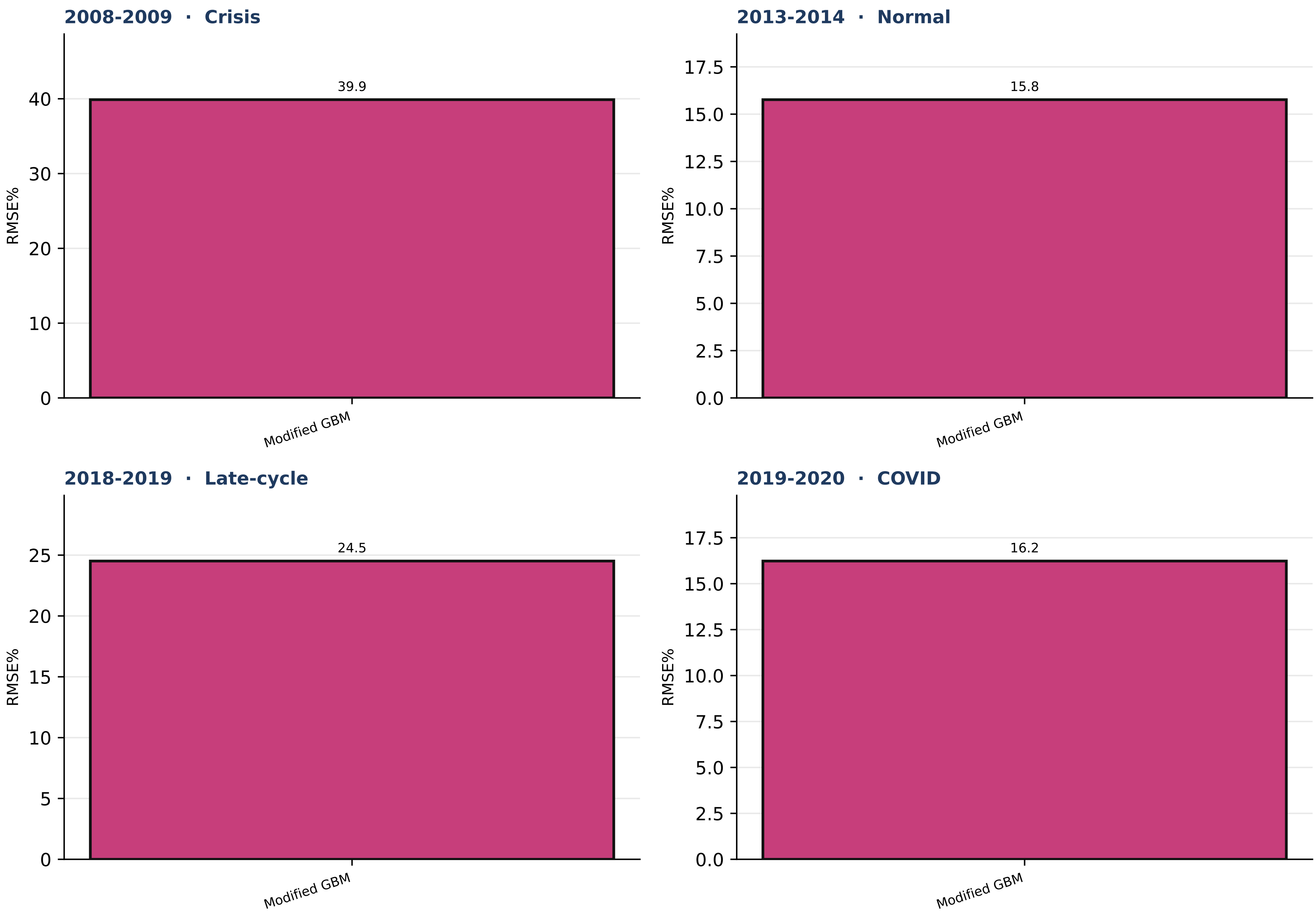
Model	Mean RMSE%	Median	# regimes best
Modified GBM	**13.36**	**12.04**	**4**

Table S4.4 · AMZN

Model	Mean RMSE%	Median	# regimes best
Modified GBM	**37.83**	**36.02**	**4**

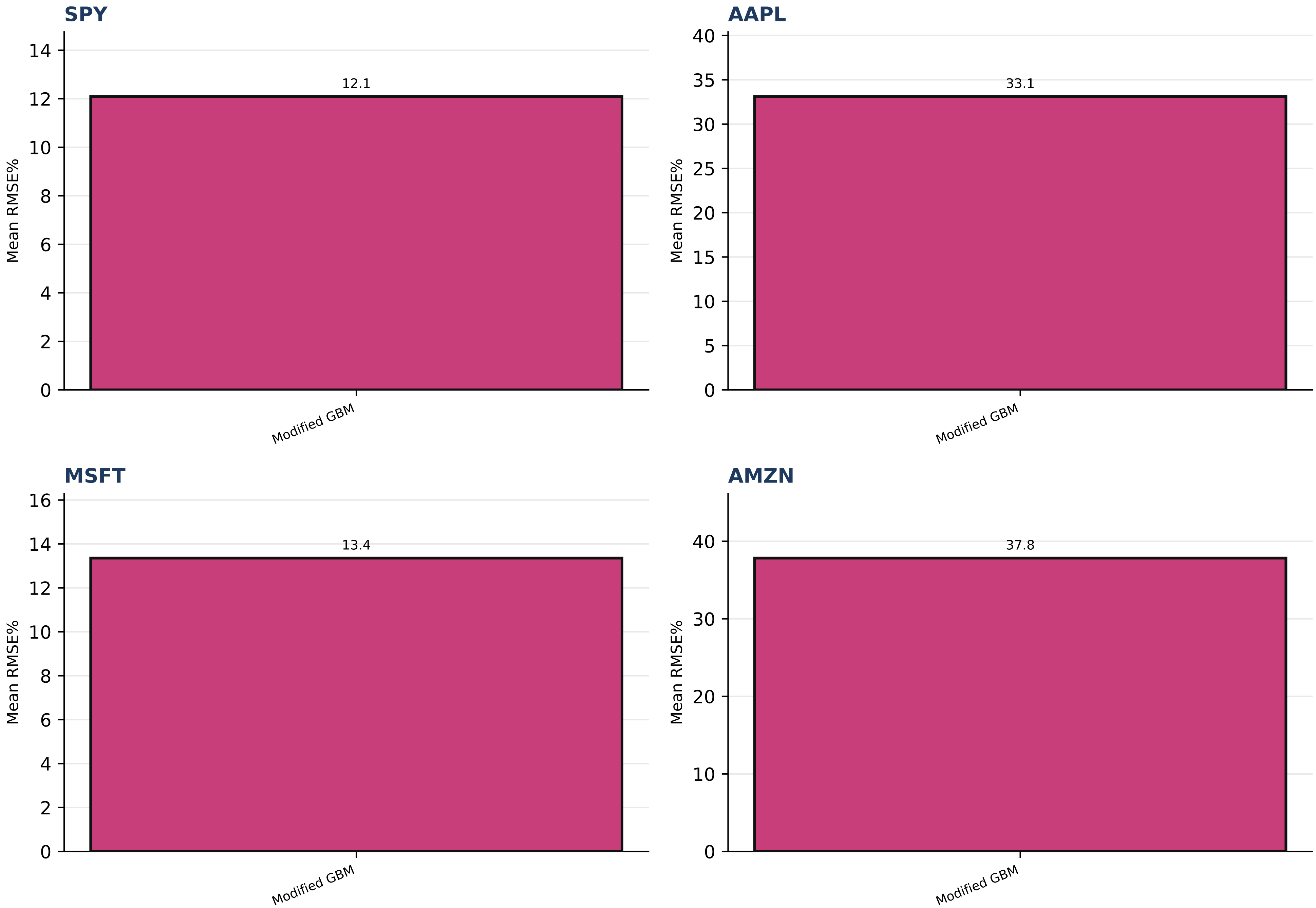
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.

Page 9 · Figure 1 · RMSE% by model and regime · mean of SPY / AAPL / MSFT / AMZN

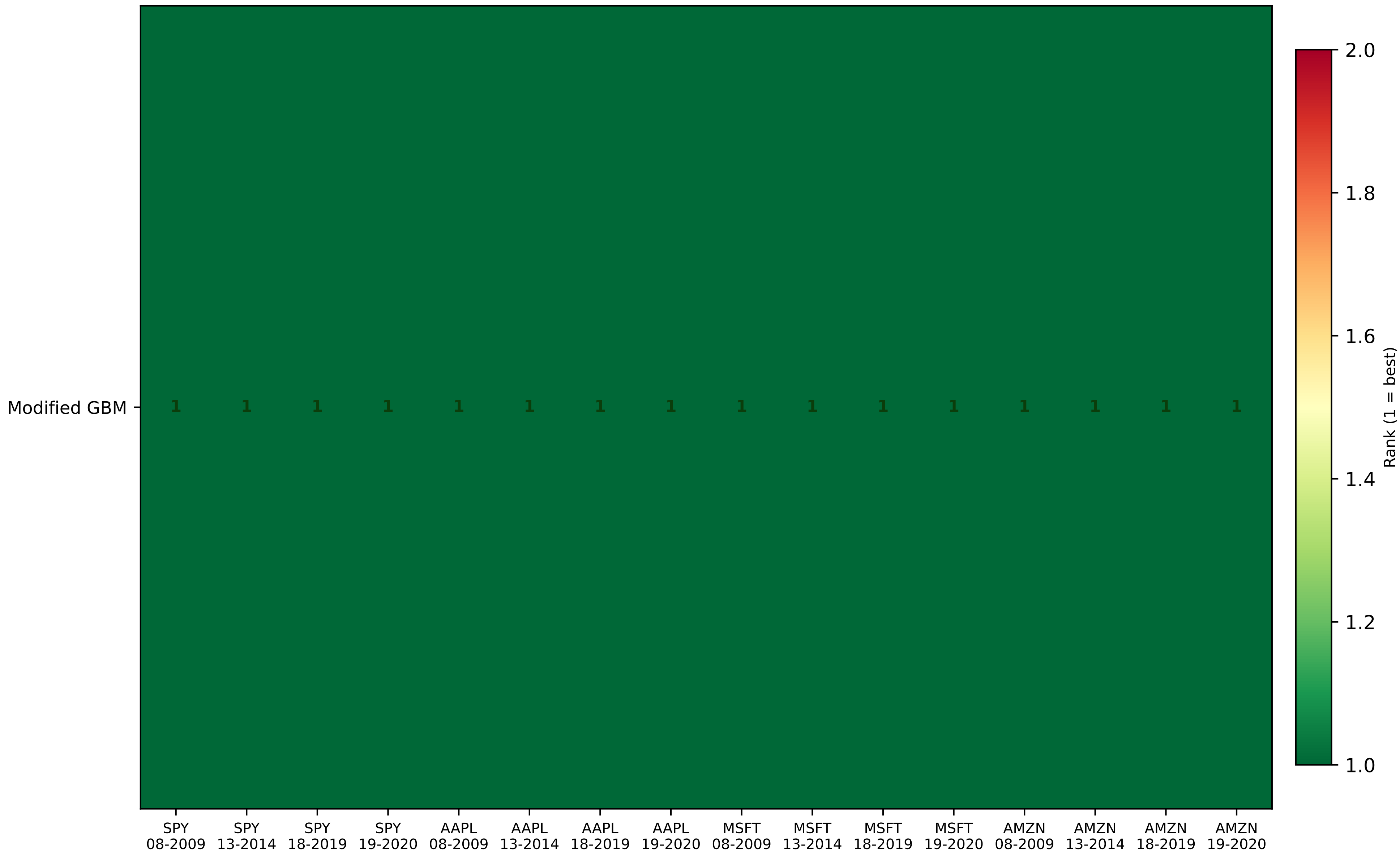


Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.

Page 10 · Figure 2 · RMSE% by model · each company (mean across four regimes)

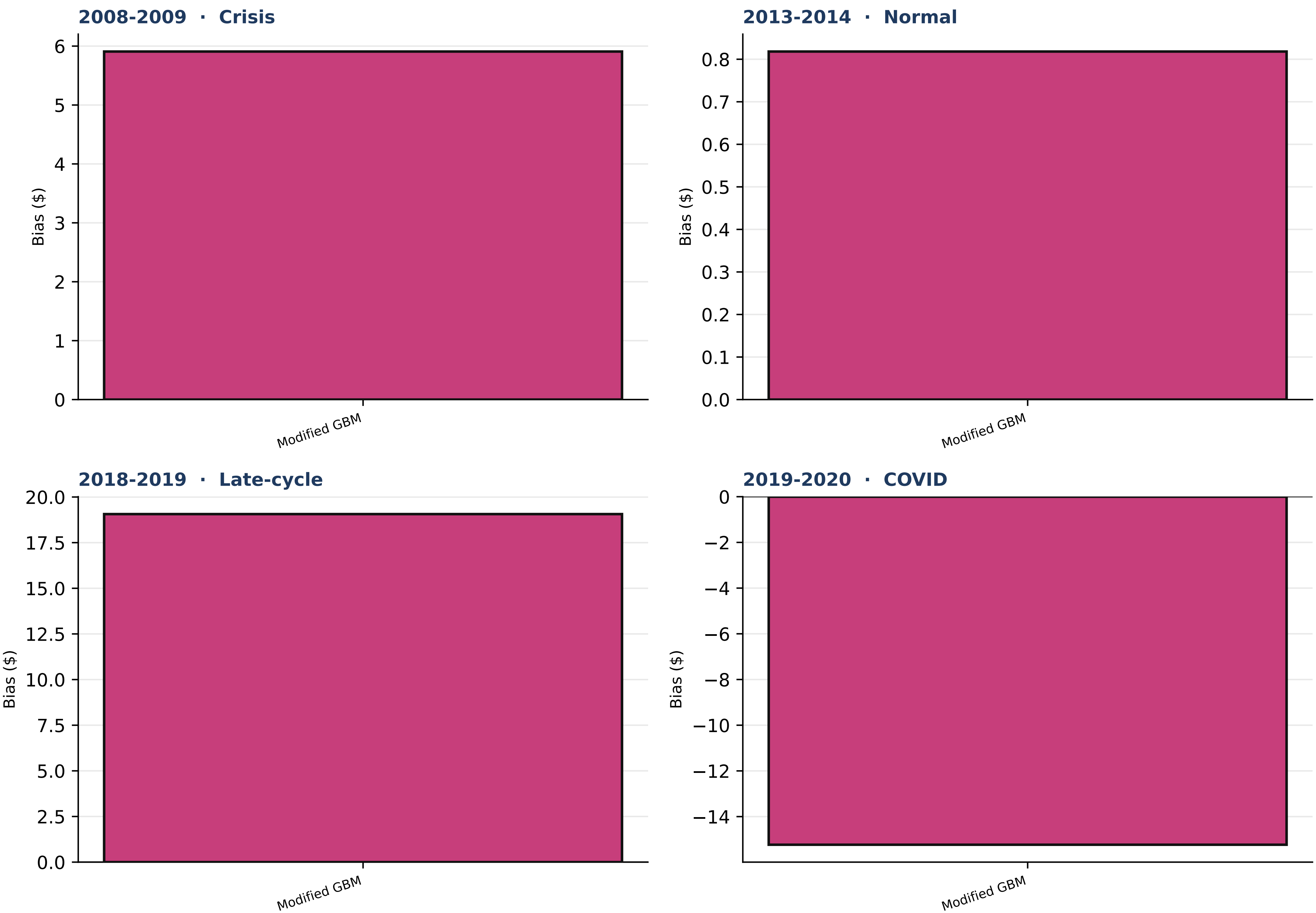


Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.

Page 12 · Figure 4 · Bias (model − market, $) · mean of four companies



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

Final analysis

Which models track realized S_t most closely?

- On mean RMSE% of the p50 path versus realized S_t across 12 company×regime cells, Modified GBM is best (mean RMSE% = 24.10; 16 of 12 cells).
- The ranking is stable: Modified GBM is best in every cell.
- These conclusions are conditional on 1.5-year monthly P-measure calibration, 10000 paths, daily steps, and the p50 scoring convention.

Do not mix these stock-path RMSE% numbers with the option-pricing RMSE% in the decision PDF.